DUALITY FOR COMPACTLY SUPPORTED PUSHFORWARD: THE LOWER SHRIEK IS LEFT ADJOINT TO THE UPPER SHRIEK IN THE PRO CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G51 (sources/stacks-duality/pinned-original-source.tex line 8721).

Allowed standard prerequisites (editorial).

- The compactification construction of the upper shriek. Same chapter, Section labelled section-upper-shriek, cited at proof lines 8762-8764. Gives the first equality.
- Coherence of the shriek and the lifting statement in the pro category. Same chapter, Lemmas labelled lemma-shriek-coherent and lemma-lift-map-plus, cited at proof lines 8768-8770. Gives the second equality.
- Adjunction for a proper morphism. Same chapter, Lemma labelled lemma-twisted-inverse-image, cited at proof lines 8772-8774. Gives the third equality.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0A9Y: sources/stacks-duality/pinned-original-source.tex, source lines 3940–3940. Same-chapter pointer into the author environment lemma at source line 3940; invoked at proof line 8750 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AU1: sources/stacks-duality/pinned-original-source.tex, source lines 4686–4686. Same-chapter pointer into the author environment lemma at source line 4686; invoked at proof line 8751 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4K: sources/stacks-duality/pinned-original-source.tex, source lines 8089–8089. Same-chapter pointer into the author environment lemma at source line 8089; invoked at proof line 8753 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Proposition 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. Then the functors $Rf_!$ and $f^!$ are adjoint in the following sense: for all $K \in D^b_{Coh}(\mathcal{O}_X)$ and $L \in D^+_{Coh}(\mathcal{O}_Y)$ we have*

$$\mathrm{Hom}_X(K, f^!L) = \mathrm{Hom}_{Pro\text{-}D^+_{Coh}(\mathcal{O}_Y)}(Rf_!K, L)$$

bifunctorially in K and L.

Proof. Choose a compactification $j : X \to \overline{X}$ over Y and denote $\overline{f} : \overline{X} \to Y$ the structure morphism. Then we have

$$\begin{aligned}
\mathrm{Hom}_X(K, f^!L) &= \mathrm{Hom}_X(K, j^*\overline{f}^!L) \\
&= \mathrm{Hom}_{\mathrm{Pro}\text{-}D^+_{Coh}(\mathcal{O}_{\overline{X}})}(Rj_!K, \overline{f}^!L) \\
&= \mathrm{Hom}_{\mathrm{Pro}\text{-}D^+_{Coh}(\mathcal{O}_Y)}(Rf_*Rj_!K, L) \\
&= \mathrm{Hom}_{\mathrm{Pro}\text{-}D^+_{Coh}(\mathcal{O}_Y)}(Rf_!K, L)
\end{aligned}$$

The first equality follows immediately from the construction of $f^!$ in Section 0A9Y. By Lemma 0AU1 we have $\overline{f}^!L$ in $D^+_{Coh}(\mathcal{O}_{\overline{X}})$ hence the second equality follows from Lemma 0G4K. Since $\overline{f}$ is proper the functor $\overline{f}^!$ is the right adjoint of pushforward by construction. This is why we have the third equality. The fourth equality holds because $Rf_! = Rf_*Rj_!$. □